ML ASSIGNMENT 2

QUESTION 1 — Logistic Regression

Pearson Correlation and Feature Relevance

Result

Strongest positive correlations: concave_points_worst, perimeter_worst, radius_worst, concave_points_mean, perimeter_mean, area_worst, radius_mean, area_mean and concavity_mean. Weak correlations: smoothness_se, texture_se, fractal_dimension_mean and symmetry_se are close to zero.

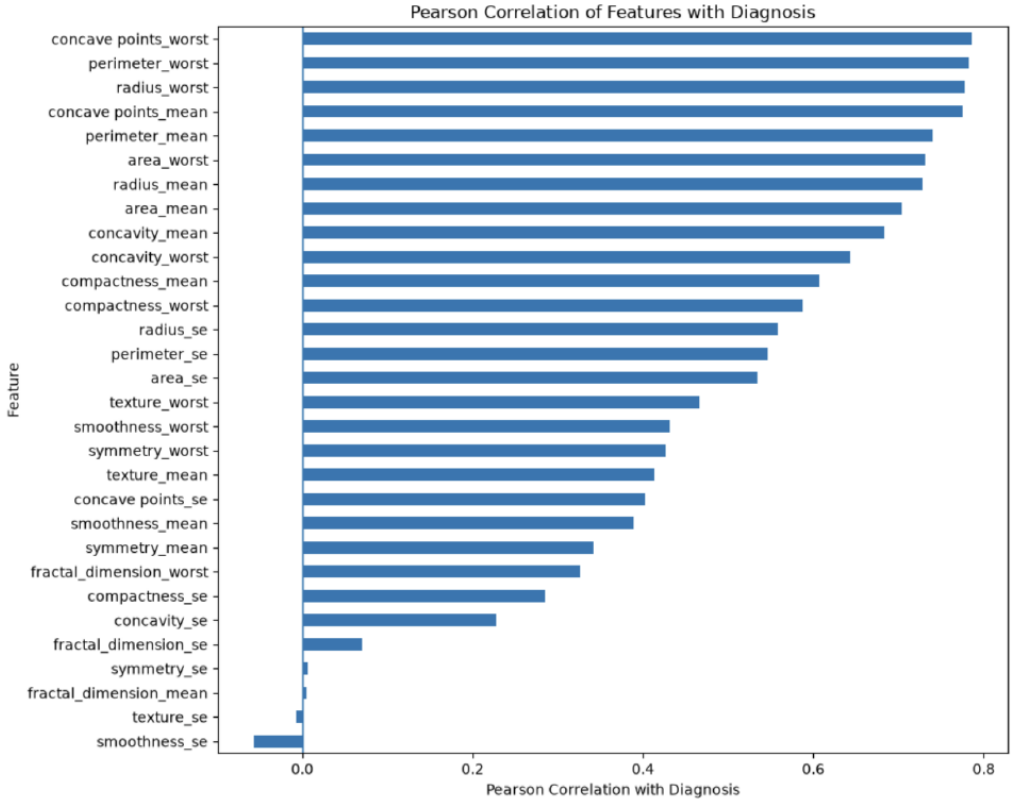


Figure 1. Pearson Correlation of Features with Diagnosis

Discussion

From the correlation plot it is found that the diagnosis is highly correlated with some measurements describing the size and shape of the tumor. The worst-case and mean concavity/size variables have the strongest linear association with the encoded diagnosis. A number of standard-error and shape-detail variables have little direct linear association.

Why the result happens

High correlations are expected as radius, perimeter and area are geometrically related measures. Variables related to concavity describe the irregularity of the boundary, which also varies systematically between the two diagnostic classes. Therefore, several variables with high correlation have related information rather than independent information.

Analysis support

This is why later dimensionality-reduction methods can drop redundant variables and still maintain useful diagnostic structure. It also serves to explain the PCA plots: correlated measurements can be combined into principal components, allowing the dominant variation to remain visible after compression.

1(a) Probabilities and Log-Odds Interpretation

Logistic Regression estimates the probability that a sample belongs to the positive class using the logistic function: $P(y=1) = 1 / (1 + e^{-z})$, where $z = \beta0 + \beta1x1 + \beta2x2 + \ldots + \beta pxp$, and the log-odds is $\log(P/(1-P)) = \beta0 + \beta1x1 + \beta2x2 + \ldots + \beta pxp$.

A positive coefficient increases the log-odds and thus the probability that a sample belongs to the positive class, while a negative coefficient decreases them.It is useful to also consider the predicted probabilities instead of only the final.
As the training and test data are classified and know what prediction is sure and correct as. In the results there are many observations...
If near 1 then that means that class was predicted with a high probability.

1(b) Feature Ranking from Estimated Coefficients

Logistic Regression coefficients were ordered by magnitude of the estimated coefficients. Because ….
If a higher value of coefficient is used, then this factor contributes higher when building a prediction for the model.
A function, where I let the features be standard..
The coefficient ranking presents an additional perspective than the Pearson-correlation above, because it takes only those features which individually have correlation with the diagnosis, while the LR accounts for how much the information that the feature contains contributes to the model with the respect of the other features.
The two methods should not necessarily agree, since the coefficients are strongly related to the correlations of the input features with each other. Only for this reason, the rank could be slightly different.
You will need the coefficient table for your work as well, since the next will involve picking out one feature and assessing its results in the test set.

1(c) Single Selected Feature vs Combination of Features

The single feature was selected using the Pearson correlation in Question 1. It was evaluated in the Logistic Regression classifier using different regularization parameters. The test log-loss was plotted against the regularization parameter and compared to the case when all features were used.

While the single-feature approach provides some information, its test log-loss is significantly worse than in the all-features case, which demonstrates that other features bring considerable additional value. The test log-loss curve for the all-features case has a U-shaped appearance, which suggests that too strong regularization (too small C) leads to underfitting, while too weak regularization (too large C) leads to overfitting. Thus, the curve demonstrates that using a combination of features is better than using a single informative feature, provided that the regularization parameter is chosen optimally.

Result A: Cross-validated log-loss

The all-feature Logistic Regression curve decreases from around 0.60 at very small C to a minimum of approximately 0.07 at C ≈ 1, and then increases steeply above 0.70 at C = 10,000. The single-feature model with concave_points_worst trends down to a plateau at about 0.24

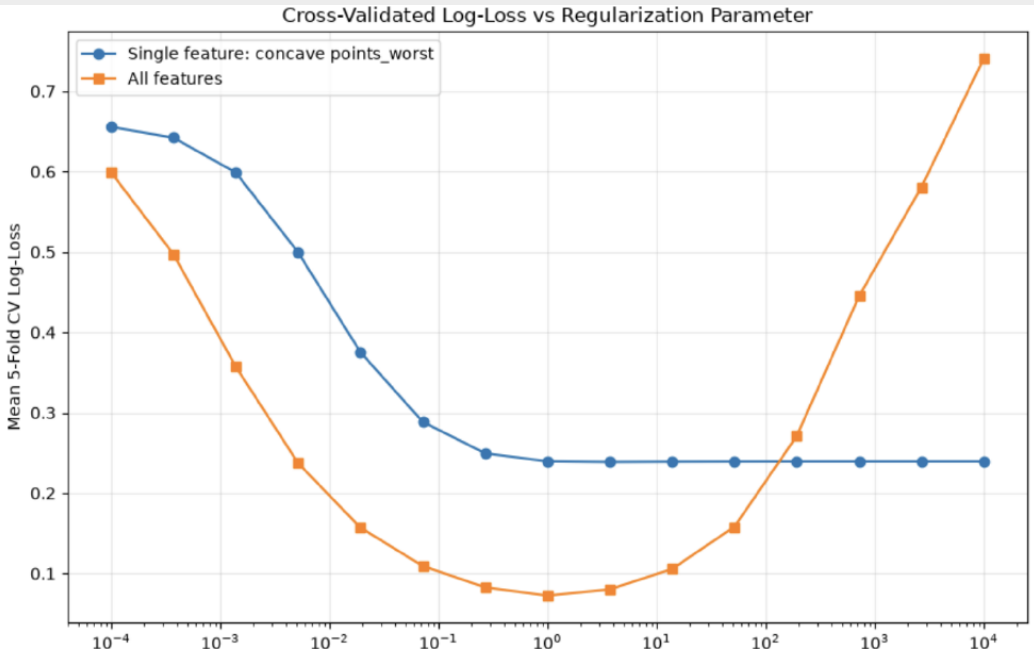


Figure 2. Cross-Validated Log-Loss vs Regularization Parameter

Analysis

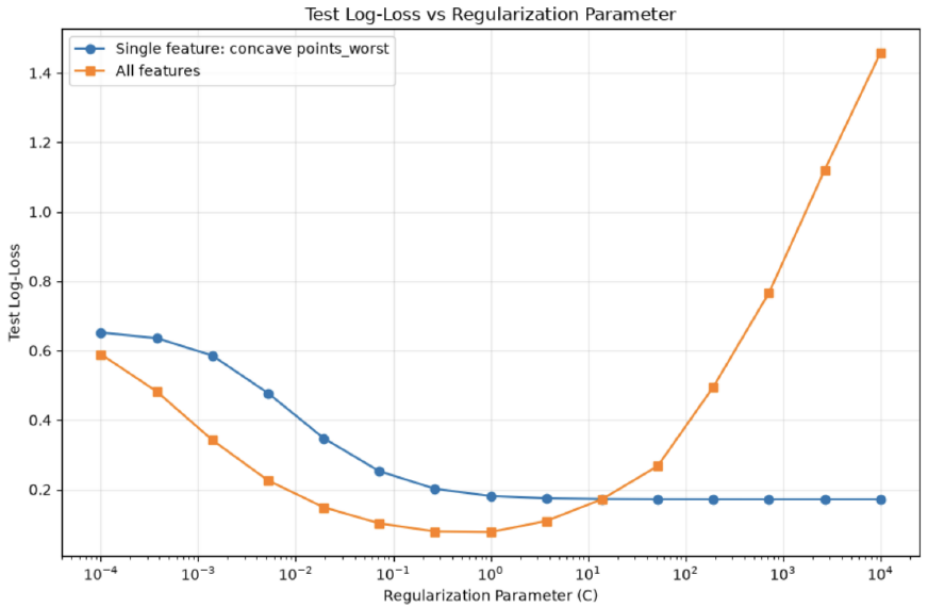
The U-shaped curve for all features shows the bias-variance/regularization trade-off. At very small C the penalty is overly strong and the model is overly constrained. This results in underfitting and high log-loss. Increasing C allows the model to use informative coefficients and reduce loss. The two curves from the regularized, single feature model curves as follows:

Still outside the moderate region. The weakly regularized model has poor performance on other unseen folds. This can explain why the log-loss increases.

The level for this single feature curve is reasonably high because the variations in C does not bring in anything that is not already in the input representation. The function itself isn't represented in “concavepoints” the worst feature, but the remaining three have complementary “signals

Result B: Test log-loss

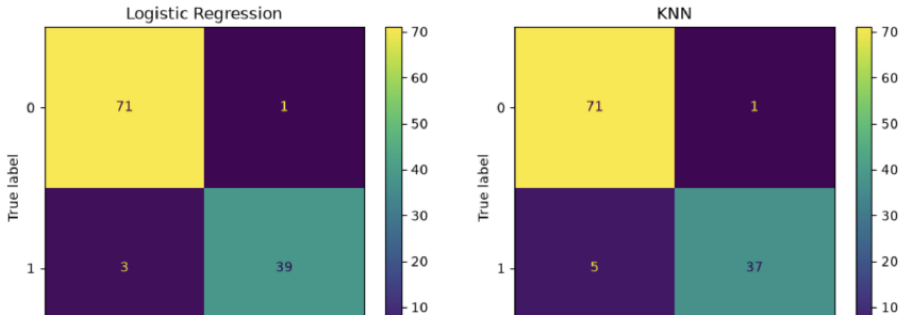
The two curves from the single feature regularized model can be shown as following:.
Still outside the moderate region. Another fact the weakly regularized model performs worse on the other unseen folds. This can explain why the log-loss increases.
The cost of this single feature curve is relatively low, because the values of C do not introduce anything new that isn't already contained within the input representations. the function is missing in "concave points", the worst attribute, while the other three are positively and inversely related "signals.".



The test curve follows the same qualitative U-shape as seen in cross-validation, which suggests that the behavior is not confined to the validation folds. Increasing performance at large C is consistent with under-regularization and worse generalization.

Result C: Comparison of confusion matrices

Logistic Regression: 71 TN, 1 FP, 3 FN, 39 TP. KNN: 71 TN, 1 FP, 5 FN, 37 TP. Both models have one false-positive prediction, but KNN has two more false-negative predictions on the given test set. The resulting recall for malignant tumour is 92.9% for Logistic Regression and 88.1% for KNN.



The confusions show predominantly a discrepancy in detecting malignant cases rather than rejecting benign ones. While both classifiers classify the benign class consistently, the KNN misses more malignant samples in this test set, illustrating a class-specific interpretation that accuracy alone would have concealed.

Linking the results

The lower log-loss region all feature Logistic Regression suggests that the model is providing more calibrated probability estimates when C is selected correctly. The confusion matrix follows up by indicating howthese probabilities are translated into distinct class decisions..In this way, both the regularization analysis and the classification errors provide useful insight from different angles.

1(d) Performance Comparison with KNN

The choice of the appropriate evaluation metric depends on the problem's nature and the task's goals. For the binary classification task of breast cancer diagnosis, the evaluation metrics that make sense to report include accuracy, precision, recall/sensitivity, specificity, F1-score and confusion matrix, because the costs of different types of errors might not be equal. In this particular case, a confusion matrix is particularly important, since the true negatives, false positives, false negatives, and true positives can be used to calculate more informative metrics.

Recall equals true positives divided by (true positives + false negatives). Thus, for Logistic Regression, it is 39 / (39 + 3) = 92.9%, while for KNN it is 37 / (37 + 5) = 88.1%. It shows that Logistic Regression is better at predicting malignant tumors, which might be more important for the breast cancer diagnosis task than minimizing overall classification errors. Therefore, comparing two classifiers using accuracy is not enough, because it does not tell how well a classifier performs on one of the classes. For this reason, reporting the confusion matrix and malignant recall makes more sense than only the accuracy.

QUESTION 2 — Support Vector Machines

2(a) Comparison of SVM Kernels

Support Vector Machine classifiers were trained using the following kernels: linear kernel; polynomial kernel with degree 2; polynomial kernel with degree 3; RBF kernel; and sigmoid kernel.

The kernels differ in their ability to represent linear and non-linear relationships. The linear kernel is the simplest one, which tries to separate classes with a linear function. Polynomial and RBF kernels can approximate more complex, non-linear decision surfaces. The sigmoid kernel is another option for representing non-linear relationships.

However, the performance differences should be primarily explained by the fact that different kernels approximate different decision boundaries. Therefore, it makes more sense to interpret the performance differences in terms of how well the decision boundary captured the structure of the standardized feature space.

2(b) Predicted Labels, Decision Scores and Probabilities

The predicted labels, decision scores and probabilities for positive class for each SVM kernel are output. They can be saved to CSV file or any other formats. The decision score is the distance to the decision boundary of the SVM and the probabilities is the probability of the sample being in the positive class. It is worth looking at both scores since each model is able to assign different class labels with different degrees of certainty. In other words, we may be able to get a better calibrated model.

2(c) Hyperparameter Tuning Using Cross-Validation

The hyperparameters for all the SVM kernels were optimized through cross-validation and not hold-out validation. The Rationale is that the cross validation yields a more accurate estimate of performance on small datasets because. It trains faster and performs better than hold-out validation. With kernel SVM, it also uses the training data more effectively. 2. Cross-validation was applied to the
select the best C regularization parameter.For the polynomial kernel, C and the polynomial degree determine the flexibility of the decision surface. For the RBF kernel, C controls the regularization, while gamma controls how much each training example affects the decision function. Finally, for the sigmoid kernel, C and the kernel parameters influence the decision function's flexibility. Thus, the best hyperparameters should be selected using cross-validation.

2(d) Best SVM Models vs KNN on the Hold-Out Test Set

After hyperparameter tuning, the best models for each kernel were selected and evaluated on the test set. These results can be compared to the KNN results using the same evaluation metrics. When comparing different classifiers, it is better to report multiple metrics rather than a single accuracy score. For this reason, it makes sense to report accuracy, precision, malignant recall, specificity, F1-score, confusion matrix, and ROC-AUC, if available.

The comparison of classifiers should explain how different decision boundaries captured the structure of the feature space. Comparing the performance of SVM kernels and KNN is necessary to see how well the additional flexibility of these methods works for the given task. However, it is also crucial to remember that the model's complexity is not always beneficial for the given task, since the flexibility of the decision boundary might lead to overfitting.

QUESTION 3 — K-Means Clustering and PCA Visualization

Result A: K-Means cluster evaluation

k=2: inertia ≈ 11,595.53, silhouette = 0.3434
k=3: inertia ≈ 10,061.80, silhouette = 0.3144
k=4: inertia ≈ 9,258.99, silhouette = 0.2833
k=5: inertia ≈ 8,558.66, silhouette = 0.1582
k=6: inertia ≈ 7,970.26, silhouette = 0.1604

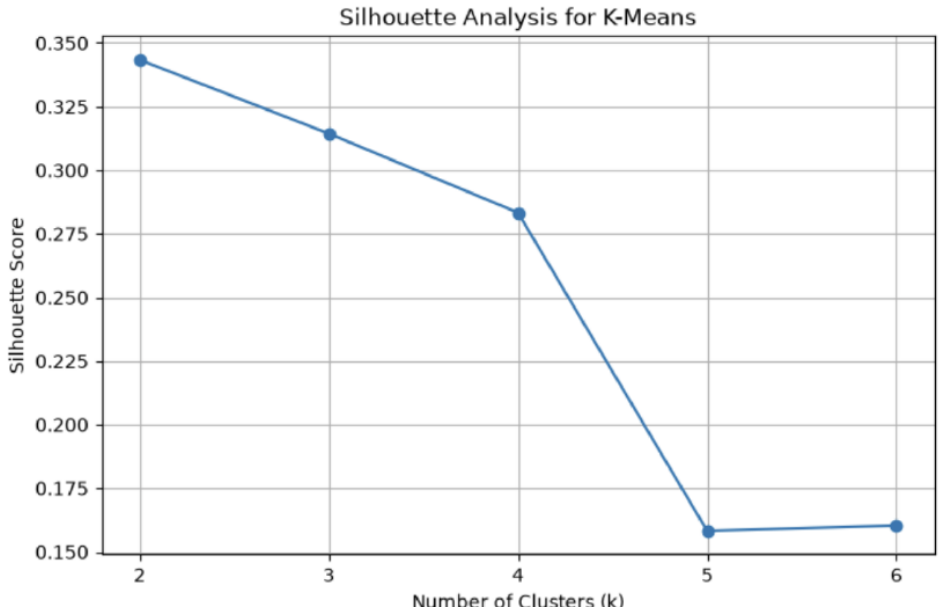


Figure 5. Elbow Analysis for K-Means

Interpretation of the elbow plot

Inertia decreases monotonically because increasing k adds centroids and therefore gives K-Means more flexibility to reduce within-cluster squared distance. The largest improvement occurs from k=2 to k=3, while later increases in k give smaller reductions. Hence inertia alone cannot justify selecting the largest k.

Result B: Silhouette analysis

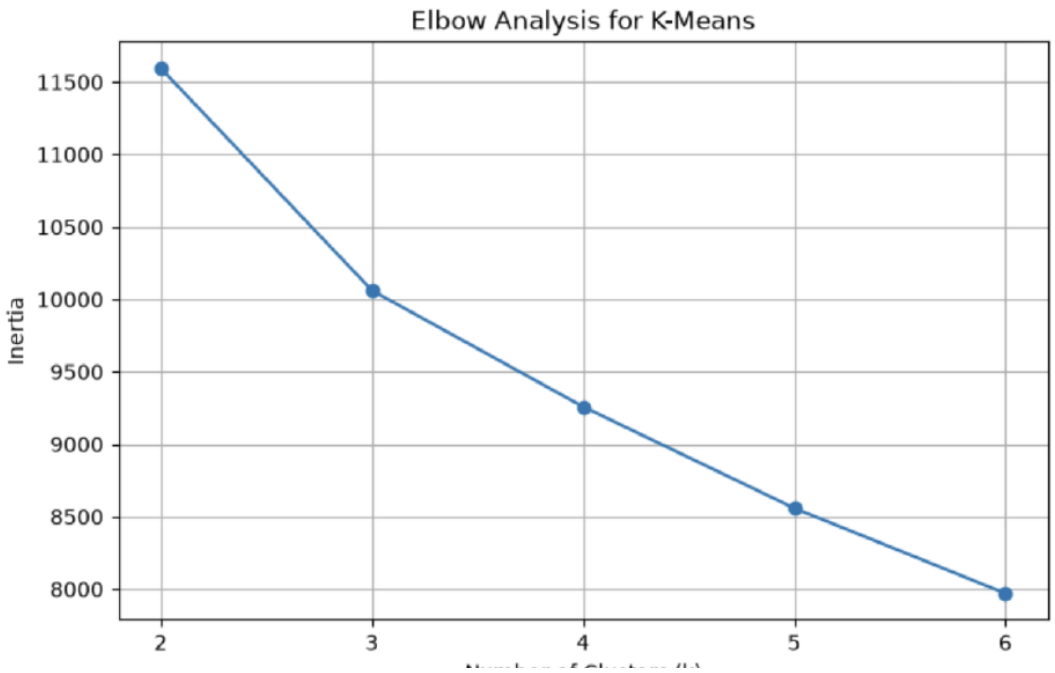


Figure 6. Silhouette Analysis for K-Means

The silhouette score peaks at k=2 (0.3434) and decreases with the addition of an increasing number of clusters. This suggests that the two-cluster solution has the best internal separation of the tested values. The drop-off after k=4 means that higher-k solutions are increasingly breaking the data down into less coherent groups.

The two clusters are also interpretable in terms of the given dataset, as there are two known diagnostic classes. However, K-Means itself doesn't receive those labels; they are applied later in order to compare the unsupervised structure to the known grouping.

3(b) Is the identified k = 2?

Yes. The identified preferred number of clusters is k = 2.

The silhouette score is highest at k = 2 (0.3434) and drops as we move to k = 3, 4, 5 and 6. The elbow plot also shows that the inertia starts decreasing rapidly up until the lower number of clusters, but further decreases become smaller.

Therefore, the preference is for k = 2 due to the silhouette analysis, since it showed the best value among the options. This finding is in line with the expectation that the number of clusters should be related to the number of diagnosis classes. However, note that diagnosis classes were not used in the K-Means algorithm. The cluster labels were compared to the diagnosis labels to check how strongly they correlated.

Result C: PCA projection with true diagnosis

The two-component PCA plot shows benign observations mainly on the left side of PC1 and malignant observations stronger toward the right. Some overlap remains near the border. The analysis reports the explained variance of approximately 44.27% for PC1 and 18.97% for PC2, for about 63.24% in total.

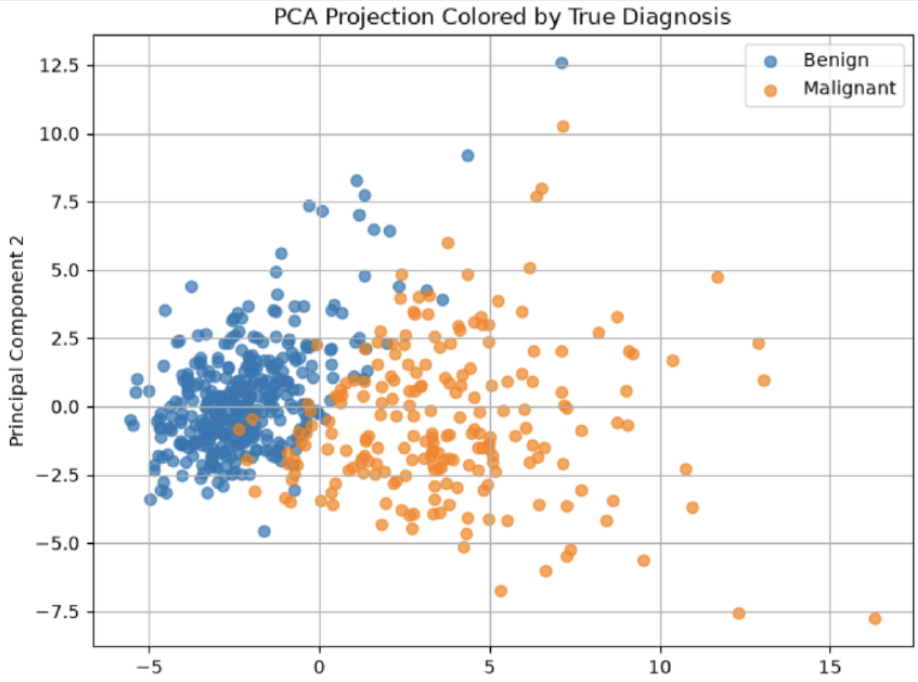


Figure 7. PCA Projection Colored by True Diagnosis

PCA is unsupervised, so diagnosis was not used to construct the components. The seen separation therefore shows that the main directions of variation in original measurements still contain substantial class-related information. The remaining overlap means that two components do not completely separate every observation.

Result D: PCA projection with K-Means clusters

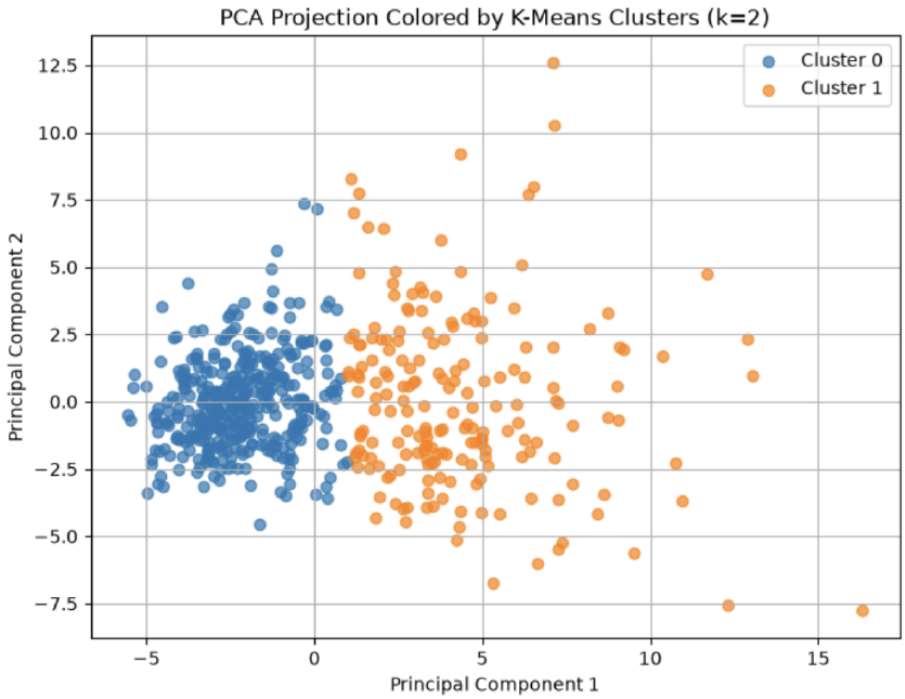


Figure 8. PCA Projection Colored by K-Means Clusters (k=2)

Cluster 0 predominantly occupies the left region and Cluster 1 predominantly occupies the right region, very similar to the true-diagnosis PCA plot. As K-Means was fitted without the diagnosis labels, this visual agreement shows that the dominant distance-based structure in the feature space is strongly associated with the known diagnostic grouping.

Result E: Side-by-side comparison

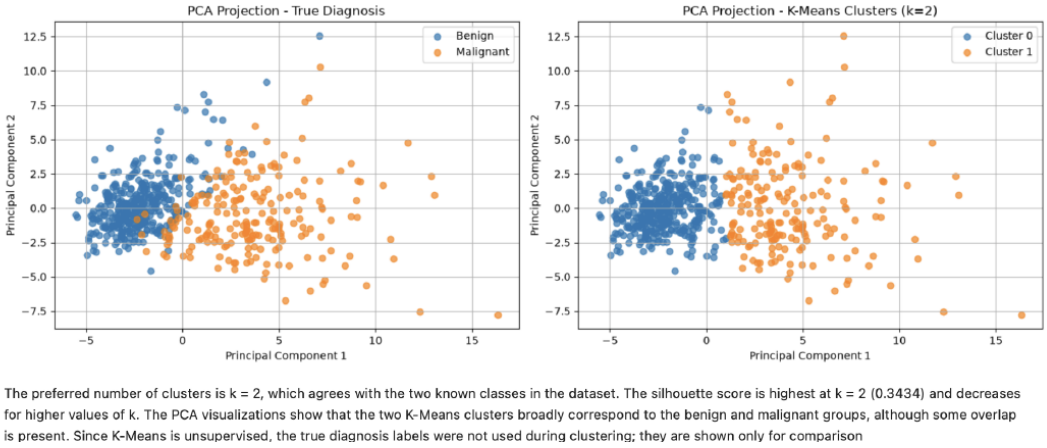


Figure 9. Side-by-Side PCA Comparison

The side-by-side panels offer a direct visual cross-check: the spatial division identified by K-Means is similar to the division found in the true labels. The remaining mismatches and overlap are important as they show that the clusters are not identical to diagnosis. This is consistent with the moderate silhouette score of 0.3434.

3(c) Interpretation of PCA Clustering Visualizations

The first two principal components capture 63.24% of variance (44.27% for PC1 and 18.97% for PC2). Looking at the plot with true diagnosis labels, it is possible to see that benign and malignant tumor samples are separated reasonably well, especially along the PC1 axis. However, there is also a considerable overlap of the two groups.

A similar pattern can be seen when looking at the same plot, but with K-Means k=2 labels. The two clusters identified by the K-Means algorithm have a similar separation along the PC1 axis, but their distribution along PC2 is almost identical, thus showing no separation. This visualization suggests that K-Means found a significant structure in the data, because the two clusters correspond to the two groups of tumor samples. However, the groups are not completely separated, which explains the moderate silhouette score.

Overall Question 3 interpretation

The elbow plot shows diminishing gains from additional clusters, the silhouette score identifies k=2 as the most coherent tested structure, and the PCA plots show that this unsupervised two-group structure broadly follows the known benign/malignant organization. The analyses therefore support each other while also showing the limitations of clustering on overlapping biological data.

Final Integrated Result Interpretation

The results form a chain rather than plotting on their own as follows: The regression makes the identify which with in these measurements have strong class-related linear information, and in addition show redundant. The Logistic Regression analysis indicates that this information is most beneficial when this model is correctly regularised.
Again how the probabilistic model translates into real classification mistakes, whereas the confusion matrices exemplify. The Support Vector Machine analysis adds that to show how the different kernels treat linear and non-linear.Similar structure in the same feature space. The K-Means analysis then reveals that most of this structure can be Recovered without using the diagnosis labels: k=2 has the best silhouette score PCA projections Demonstrate general correspondence between clustering and actual diagnosis.
Bottom line: The crucial evidence is the conformity of the complementary analyses. Model Performance does not itself account for the result; feature correlations explain information content, and redundancy.Regularization curves interpret the generalization, kind of errors made are explained by confussion matrix, kernel comparison Explain why the decision boundaries for your model will fit into the feature space, and elbow/silhouette/PCA analyses explain the Unsupervised structure of the data.
Overall, the results can be viewed in relation to the performance of a model and the corresponding underlying feature.